Epilepsy in infancy
Indian data: the second commonest genetic cause of infantile spasms is treatable with vitamin B6

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the doctor's.

Indian study: Nagarajan B, Gowda VK, Yoganathan S, ... Sankhyan N, Sahu JK. *Landscape of genetic infantile epileptic spasms syndrome: A multicenter cohort of 124 children from India.* Epilepsia Open 2023;8(4):1383-1404. doi:10.1002/epi4.12811. Six centres in India.

Who was studied

- 124 children with infantile epileptic spasms syndrome and a confirmed genetic cause, enrolled January 2021 to June 2022
- 60% boys; consanguinity in 15%
- Median age at spasm onset 5 months; median age at genetic diagnosis 12 months
- Developmental delay before spasms in 94.3%; spasms before 6 months in 86.2%

What was found

- 105 single-gene disorders, 19 chromosomal disorders; 51 genes, 92 variants (25 novel)
- Commonest causes: trisomy 21 (14), **ALDH7A1 (10)**, SCN2A (7), CDKL5 (6), ALG13 (5); KCNQ2, STXBP1, SCN1A, NTRK2, WWOX (4 each)
- About 60% responded initially to hormonal therapy

Why a genetic cause matters

Some causes have specific treatment: **ALDH7A1** causes pyridoxine-dependent epilepsy, treated with vitamin B6. For others, the result guides medicine choice: in **SCN1A**-related epilepsy, some sodium-channel-blocking seizure medicines can worsen seizures (GeneReviews). A diagnosis also gives the family a recurrence risk for future pregnancies and ends a long search for the cause.

Testing guideline

- Genetic testing is strongly recommended for every person with unexplained epilepsy, at any age (NSGC practice guideline).
- First-tier: exome or genome sequencing, or a multi-gene epilepsy panel of more than 25 genes; then chromosomal microarray. Exome or genome is conditionally preferred over a panel.
- Pre- and post-test genetic counselling; prenatal testing in a later pregnancy is for disease only, fetal sex never disclosed (PCPNDT).

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Epilepsy in infancy: how to approach doctors

Topic	What to do or say
Doctors to visit	Paediatric neurologists (key), paediatricians, neonatologists (seizures in NICU), epilepsy centres.
Which children	Seizures starting before age 1; delayed development; seizures not controlled by medicines; MRI and metabolic tests found no cause; similar illness in the family.
Opening question	"Doctor, do you test infants whose seizures have no clear cause?"
Fact to quote	In 124 Indian children with genetic infantile spasms, the second commonest cause, ALDH7A1, is treatable with vitamin B6 (Nagarajan et al. 2023).
Test to suggest	Exome or genome, or an epilepsy panel of more than 25 genes, first; microarray next (NSGC guideline). Trio if possible.
"MRI and EEG are done"	"They show where the problem is; a genetic test looks for why."
"Reports take time"	"Treatment continues; the test runs alongside." Confirm report time with the lab before quoting.
Sample	EDTA blood; parents' samples for trio; counselling and written consent first.
Never say	That the test will stop the seizures or guarantees an answer. Treatment decisions are the doctor's.

References

- Nagarajan B et al. Landscape of genetic infantile epileptic spasms syndrome: a multicenter cohort of 124 children from India. Epilepsia Open 2023;8(4):1383-1404. doi.org/10.1002/epi4.12811
- Smith L, Malinowski J, Ceulemans S, et al. Genetic testing and counseling for the unexplained epilepsies: an evidence-based practice guideline of the National Society of Genetic Counselors. J Genet Couns 2023. nsgc.org
- GeneReviews: Pyridoxine-dependent epilepsy (ALDH7A1); SCN1A seizure disorders.

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